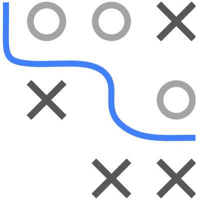
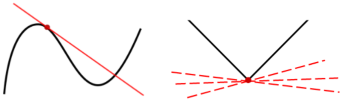
Optimization in Machine Learning



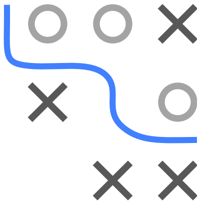
Foundations
Differentiation and Derivatives



Learning goals

- Uni- & multivariate differentiation
- Gradient, partial derivatives
- Jacobian matrix
- Hessian matrix
- Lipschitz continuity

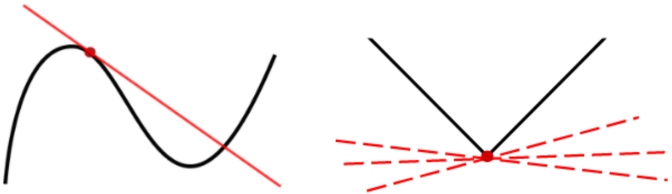
UNIVARIATE DIFFERENTIABILITY



Definition: A function $f : \mathcal{S} \subseteq \mathbb{R} \to \mathbb{R}$ is said to be **differentiable** for each inner point $x \in \mathcal{S}$ if the following limit exists:

$$f'(x) := \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

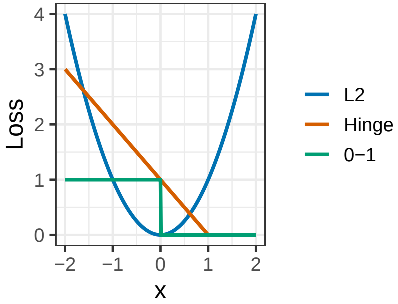
Intuitively: f can be approximated locally by a lin. fun. with slope $m = f'(x)$.



Left: Function is differentiable everywhere. **Right:** Not differentiable at the red point.

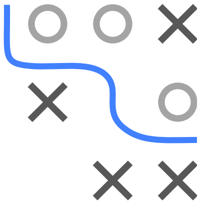
SMOOTH VS. NON-SMOOTH

- **Smoothness** of a function $f : \mathcal{S} \to \mathbb{R}$ is measured by the number of its continuous derivatives
- $\mathcal{C}^k$ is class of k-times continuously differentiable functions ($f \in \mathcal{C}^k$ means $f^{(k)}$ exists and is continuous)
- In this lecture, we call f “smooth”, if at least $f \in \mathcal{C}^1$



f_1 is smooth, f_2 is continuous but not differentiable, and f_3 is non-continuous.

MULTIVARIATE DIFFERENTIABILITY

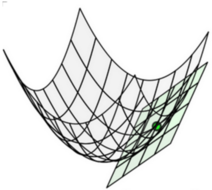


Definition: For a function $f : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}$ of d variables $x_1, \dots, x_d$, **partial derivatives** are defined as

$$\frac{\partial f}{\partial x_1} = \lim_{h \to 0} \frac{f(x_1 + h, x_2, \dots, x_d) - f(\boldsymbol{x})}{h}$$

$$\vdots$$

$$\frac{\partial f}{\partial x_d} = \lim_{h \to 0} \frac{f(x_1, \dots, x_{d-1}, x_d + h) - f(\boldsymbol{x})}{h}$$



Geometrically: Similarly to the 1D case, the vector of partial derivatives can be used to determine a tangent hyperplane. ▸ Click for source

GRADIENT

- Specifically, the vector of partial derivatives is called the **gradient**:

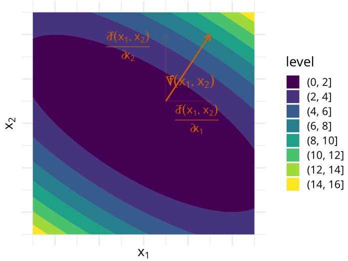
$$\nabla_{\mathbf{x}} f \text{ or } \nabla f := \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_d} \right)$$

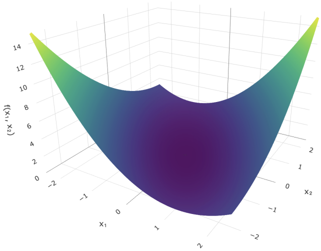
  note that this is a row vector!

- This gradient of f can be used to linearly approximate f:

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \nabla_{\mathbf{x}} f(\mathbf{x}) \mathbf{h} + o(\mathbf{h})$$

- **Example:** $f(\mathbf{x}) = x_1^2/2 + x_1 x_2 + x_2^2 \Rightarrow \nabla f(\mathbf{x}) = (x_1 + x_2, x_1 + 2x_2)$





DIRECTIONAL DERIVATIVE

The **directional derivative** tells how fast $f : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}$ is changing w.r.t. an arbitrary direction $\mathbf{v}$:

$$D_{\mathbf{v}} f(\mathbf{x}) := \lim_{h \to 0} \frac{f(\mathbf{x} + h\mathbf{v}) - f(\mathbf{x})}{h} = \nabla f(\mathbf{x}) \mathbf{v}$$

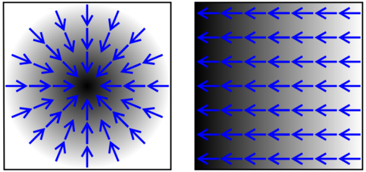
Example: The directional derivative for $\mathbf{v} = (1, 1)$ is:

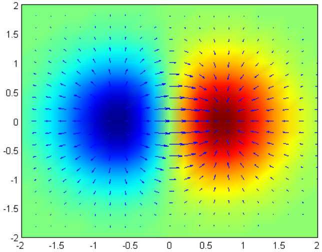
$$\begin{aligned} D_{\mathbf{v}} f(\mathbf{x}) &= \nabla f(\mathbf{x}) \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{\partial f}{\partial x_1} + \frac{\partial f}{\partial x_2} \end{aligned}$$

NB: Some people require that $||\mathbf{v}|| = 1$. Then, we can identify $D_{\mathbf{v}} f(\mathbf{x})$ with the instantaneous rate of change in direction $\mathbf{v}$, i.e. $\lim_{h \to 0} \frac{f(\mathbf{x}+h\mathbf{v})-f(\mathbf{x})}{h}$ – and in our example we would have to divide by $\sqrt{2}$.

IMPORTANT PROPERTIES OF THE GRADIENT

- **Orthogonal** to level curves/surfaces of a function
- Points in direction of **greatest increase** of f





IMPORTANT PROPERTIES OF THE GRADIENT

- To prove 1, consider that a *level curve* of a function $f : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}$ is any parametrized, differentiable curve (see *implicit function theorem*) $\boldsymbol{x}(t)$, $t \in \mathcal{T} \subseteq \mathbb{R}_{\geq 0}$ so that $f(\boldsymbol{x}(t)) = c$ for some constant c.
- **Proof of 1.**: Consider a point $\boldsymbol{x}_0 \in \mathbb{R}^d$ so that $f(\boldsymbol{x}_0) = c$ and $\exists t_0 \in \mathcal{T} : \boldsymbol{x}(t_0) = \boldsymbol{x}_0$. To prove orthogonality, we want to show $\langle \nabla_{\boldsymbol{x}} f(\boldsymbol{x}_0), \frac{d\boldsymbol{x}}{dt}(t_0) \rangle = 0$.
- Note that

$$\left\langle \nabla_{\boldsymbol{x}} f(\boldsymbol{x}_0), \frac{d\boldsymbol{x}}{dt}(t_0) \right\rangle = \nabla_{\boldsymbol{x}} f(\boldsymbol{x}(t_0)) \frac{d\boldsymbol{x}}{dt}(t_0) = \frac{d(f \circ \boldsymbol{x})}{dt}(t_0)$$

  By the chain rule, see slide-set “Matrix Calculus”.

- Since $f \circ \boldsymbol{x}$ is a constant function in t, the derivative will always equal zero and so $\frac{d(f \circ \boldsymbol{x})}{dt}(t_0) = 0$, proving the statement.

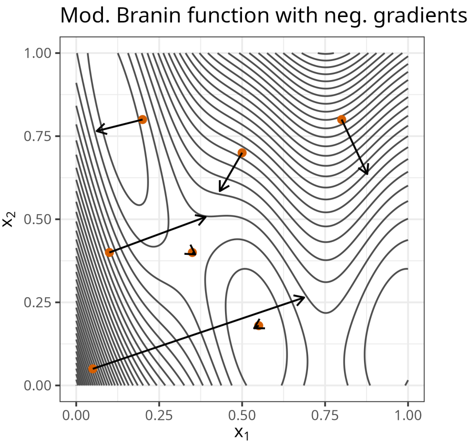
IMPORTANT PROPERTIES OF THE GRADIENT

- **Proof of 2.**: Let $\mathbf{v}$ be a vector with $\|\mathbf{v}\| = 1$ and θ the angle between $\mathbf{v}$ and $\nabla f(\mathbf{x})$.

$$D_{\mathbf{v}} f(\mathbf{x}) = \nabla f(\mathbf{x}) \mathbf{v} = ||\nabla f(\mathbf{x})|| \, ||\mathbf{v}|| \cos(\theta) = ||\nabla f(\mathbf{x})|| \cos(\theta)$$

  by the cosine formula for dot products and $\|\mathbf{v}\| = 1$.
- $\cos(\theta)$ is maximal if $\theta = 0$, hence if $\mathbf{v}$ and $\nabla f(\mathbf{x})$ point in the same direction.
- (Alternative proof: Apply Cauchy-Schwarz to $\nabla f(\mathbf{x}) \mathbf{v}$ and look for equality.)
- Analogous: Negative gradient $-\nabla f(\mathbf{x})$ points in direction of greatest *de*crease

IMPORTANT PROPERTIES OF THE GRADIENT



Length of arrows is norm of their gradient.

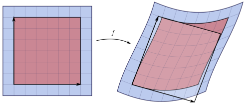
JACOBIAN MATRIX

For vector-valued function $f : \mathcal{S} \subseteq \mathbb{R}^d \longrightarrow \mathbb{R}^m, \boldsymbol{x} \mapsto (f_1(\boldsymbol{x}), \ldots, f_m(\boldsymbol{x}))^T$, $f_j : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}$, the **Jacobian** matrix $J_f : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}^{m \times d}$ generalizes gradient by placing all ∇f_j in its rows:

$$J_f(\boldsymbol{x}) \text{ or } \nabla f(\boldsymbol{x}) = \begin{pmatrix} \nabla f_1(\boldsymbol{x}) \\ \vdots \\ \nabla f_m(\boldsymbol{x}) \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1(\boldsymbol{x})}{\partial x_1} & \cdots & \frac{\partial f_1(\boldsymbol{x})}{\partial x_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(\boldsymbol{x})}{\partial x_1} & \cdots & \frac{\partial f_m(\boldsymbol{x})}{\partial x_d} \end{pmatrix}$$

We will mainly use the ∇f notation.

- Jacobian gives best linear approximation of distorted volumes



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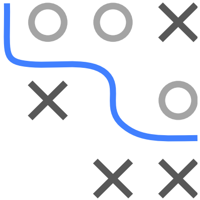
JACOBIAN DETERMINANT

- Let $f \in \mathcal{C}^1$ and $\boldsymbol{x}_0 \in \mathcal{S} \subseteq \mathbb{R}^d$.
- **Inverse function theorem:** Let $\boldsymbol{y}_0 = f(\boldsymbol{x}_0)$. If $\det(J_f(\boldsymbol{x}_0)) \neq 0$, then
  1. f is invertible in a neighborhood of $\boldsymbol{x}_0$,
  2. $f^{-1} \in \mathcal{C}^1$ with $J_{f^{-1}}(\boldsymbol{y}_0) = J_f(\boldsymbol{x}_0)^{-1}$.
- $|\det(J_f(\boldsymbol{x}_0))|$: factor by which f expands/shrinks volumes near $\boldsymbol{x}_0$
- If $\det(J_f(\boldsymbol{x}_0)) > 0$, f preserves orientation near $\boldsymbol{x}_0$
- If $\det(J_f(\boldsymbol{x}_0)) < 0$, f reverses orientation near $\boldsymbol{x}_0$

HESSIAN MATRIX



For real-valued function $f : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}$, the **Hessian** matrix $\nabla^2 : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}^{d \times d}$ contains all its second derivatives (if they exist):

$$\nabla^2 f(\boldsymbol{x}) = \left(\frac{\partial^2 f(\boldsymbol{x})}{\partial x_i \partial x_j} \right)_{i,j=1,\dots,d}$$

Note: Hessian of f is Jacobian of ∇f. Also, the Hessian is often denoted by $H(\boldsymbol{x}) \hat{=} \nabla^2 f(\boldsymbol{x})$. **Example**: Let $f(\boldsymbol{x}) = \sin(x_1) \cdot \cos(2x_2)$.

Then:

$$\nabla^2 f(\boldsymbol{x}) = \begin{pmatrix} -\cos(2x_2) \cdot \sin(x_1) & -2\cos(x_1) \cdot \sin(2x_2) \\ -2\cos(x_1) \cdot \sin(2x_2) & -4\cos(2x_2) \cdot \sin(x_1) \end{pmatrix}$$

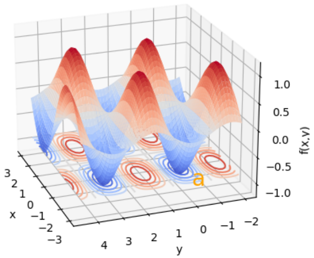
- If $f \in \mathcal{C}^2$, then $\nabla^2 f$ is symmetric
- Many local properties (geometry, convexity, critical points) are encoded by the Hessian and its spectrum ($\to$ later)

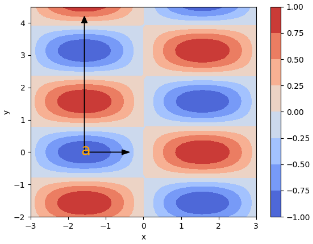
LOCAL CURVATURE BY HESSIAN

Eigenvector corresponding to largest (resp. smallest) **eigenvalue** of Hessian points in direction of largest (resp. smallest) **curvature**
Example (previous slide)**:** For $\boldsymbol{a} = (-\pi/2, 0)^T$, we have

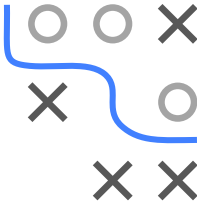
$$\nabla^2 f(\boldsymbol{a}) = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$$

and thus $\lambda_1 = 4, \lambda_2 = 1$, $\boldsymbol{v}_1 = (0, 1)^T$, and $\boldsymbol{v}_2 = (1, 0)^T$.





LIPSCHITZ CONTINUITY



Function $h : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}^m$ is **Lipschitz continuous** if slopes are bounded:

$$||h(\boldsymbol{x}) - h(\boldsymbol{y})|| \leq L||\boldsymbol{x} - \boldsymbol{y}|| \quad \text{for each } \boldsymbol{x}, \boldsymbol{y} \in \mathcal{S} \subseteq \mathbb{R}^d \text{ and some } L > 0$$

- **Examples** ($d = m = 1$)**:** $\sin(x)$, $|x|$
- **Not** examples: $1/x$ (but *locally* Lipschitz continuous), $\sqrt{x}$
- If $m = d$ and h **differentiable**:

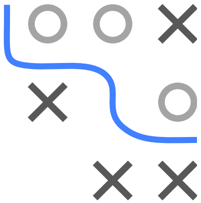
$$h \text{ Lipschitz continuous with constant } L \iff J_h \preccurlyeq L \cdot \boldsymbol{I}_d$$

Note: $\mathbf{A} \preccurlyeq \boldsymbol{B} :\iff \boldsymbol{B} - \mathbf{A}$ is positive semidefinite, i.e., $\boldsymbol{v}^T(\boldsymbol{B} - \mathbf{A})\boldsymbol{v} \geq 0 \;\; \forall \boldsymbol{v} \neq 0$. **Proof** of "$\Rightarrow$" for $d = m = 1$**:**

$$h'(x) = \lim_{\epsilon \to 0} \frac{h(x+\epsilon) - h(x)}{\epsilon} \leq \lim_{\epsilon \to 0} \left| \frac{h(x+\epsilon) - h(x)}{\epsilon} \right| \leq \lim_{\epsilon \to 0} L = L$$

[**Proof** of "$\Leftarrow$" by mean value theorem: Show that $\lambda_{\max}(J_h) \leq L$.]

LIPSCHITZ GRADIENTS / *L*-SMOOTHNESS



- f is called L**-smooth** (or has L**-Lipschitz gradients**) if

$$||\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})|| \leq L||\boldsymbol{x} - \boldsymbol{y}|| \qquad \forall \boldsymbol{x}, \boldsymbol{y} \in \mathcal{S}$$

- For $f \in \mathcal{C}^2$, since $\nabla^2 f$ is the Jacobian of ∇f and symmetric:

$$f \text{ is } L\text{-smooth} \iff -L\boldsymbol{I}_d \preccurlyeq \nabla^2 f(\boldsymbol{x}) \preccurlyeq L \cdot \boldsymbol{I}_d \quad \forall \boldsymbol{x}$$

- Equivalently, all eigenvalues of $\nabla^2 f$ have magnitude bounded by L
- **Interpretation:** Curvature in any direction is bounded by L
- Lipschitz gradients occur frequently in machine learning $\implies$ Fairly **weak assumption**

DESCENT LEMMA

- For L-smooth f: integrating the Lipschitz bound on ∇f yields a **quadratic upper bound**:

$$f(\boldsymbol{y}) \leq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})(\boldsymbol{y} - \boldsymbol{x}) + \frac{L}{2}||\boldsymbol{y} - \boldsymbol{x}||^2 \qquad \forall \boldsymbol{x}, \boldsymbol{y}$$

- Geometric reading: f is upper-bounded by a quadratic that touches f at $\boldsymbol{x}$
- Cornerstone for analyzing **gradient descent** (later: pick step size $1/L$, guaranteed progress per step)
- Counterpart of the lower bound for strongly convex f (see slide-set “Convexity”)